



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Production Engg.
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: TY Final Year B.Tech.

Course: Operations Research

Course Code: PCC-PE305

Date: 14.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

*Registration No: - _____

Note:

- Assume suitable data whenever necessary and state it clearly.
- Solve **any TWO** questions from questions 1 to 3.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.1	Solve the following problem using the Simplex method Minimize $Z = x_1 + 3x_2 + 3x_3$, Subject to, $3x_1 - x_2 + 2x_3 \leq 7$, $2x_1 + 4x_2 \geq -12$, $-4x_1 + 3x_2 + 8x_3 \leq 10$, Where, $x_1, x_2, x_3 \geq 0$.	10	CO1	BT3
Q.2	Food X contains 6 units of vitamin A per gram and 7 units of vitamin B per gram and costs 12 paise per gram. Food Y contains 8 units of vitamin A per gram and 12 units of vitamin B per gram and costs 20 paise per gram. The daily minimum requirement of vitamin A and vitamin B is 100 units and 120 units respectively. Find the minimum cost of product mix by using Big-M method.	10	CO1	BT4
Q.3	Solve the following problem using the Two-phase method, Maximize $Z = 5x_1 - 4x_2 + 3x_3$, Subject to, $2x_1 + x_2 - 6x_3 = 20$, $6x_1 + 5x_2 + 10x_3 \leq 76$, $8x_1 - 3x_2 + 6x_3 \leq 50$, Where, $x_1, x_2, x_3 \geq 0$.	05	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Conceptualize and formulate various real-life problems.
CO2	Learn various analytical techniques to evaluate the problems.
CO3	Use software tool for solving LPP.
CO4	Generate optimal solutions for various queuing, transportation, assignment and dynamic programming problems



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Department of Production Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester I)

Class: TY B.Tech.

Course: Smart Materials

Course Code: PEC-PE305

Date: 15.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Questions	Marks	CO	BT
Q.1	A Explain in detail the Piezoelectric Materials along with properties, governing relations, strength and weaknesses.	5	CO3	BT2
	B Illustrate how Shape Memory Alloys works, along with its principal, properties and use.	5	CO3	BT1
Q.2	A Illustrate the synthesis techniques of Carbon Nanotubes and Carbon nanostructures and demonstrate their applications in engineering systems.	4	CO4	BT3
	B Analyse the working principle of Thermoelectric Materials and Peltier devices and differentiate their strengths and weaknesses for cooling and energy conversion applications.	6	CO4	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Classify and select different types of smart materials.
CO 2	Comprehend Important Concepts and principles of Smart Materials.
CO 3	Synthesis, sensing and actuation of Piezoelectric Materials, Magneto strictive Materials, Shape Memory Alloys, Electroactive Polymers
CO 4	Synthesis, sensing and actuation of Ferrofluids and Magneto rheological Fluids, Soft Matter, Carbon Nanotubes and Carbon nanostructures, Thermoelectric Materials
CO 5	Classify and select Smart Materials for Energy Applications: Materials used for energy storage.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: TY B.Tech.

Course : Industrial Engineering

Course Code: PEC-PE301

Date: 14.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Question 1 is compulsory and solve any three questions in Question 2.**

Q.N.			Marks	CO	BT
Q.1		What are the equipments used for material handling in process layout?	02	CO3	BT1
Q.2	a	State the objectives of micromotion study and state the meaning and use of Therblig.	06	CO2	BT2
	b	Explain the importance of principle of motion economy and state the principles of motion economy related to the use of human body.	06	CO2	BT3
	C	A work sampling study was conducted for 30 hours. During this time, 1000 observations were made, and 850 of them showed the worker was working. The total number of articles produced during the study period were 320. The ratio between manual and machine elements was 2:1. The worker's average performance rating was 105%, and a 10% allowance for delays was added to the standard time. Calculate the standard time per piece.	06	CO2	BT5
	d	A company manufactures three products R, S and T using the same manufacturing facilities arranged in a six departments. The material handling is done by a forklift. The containers can carry 250, 500 and 750 pieces of the products R, S and T respectively. The annual demand for each product is 15000 units. Sequence of operations of product movement are given below. Construct the Travel chart. Product Departments R A → B → E → D → C → F S A → C → B → D → E → F T C → A → E → B → D → F	06	CO3	BT4

== ALL THE BEST ==



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering.
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Renewable Energy
Resources

Course Code: OE-PE301

Date: 17.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Questions.	Marks	CO	BT
Q.1	a	With the help of a neat sketch, discuss the different types of rotors used in wind turbines.	06	CO3	BT2,BT3
	b	Discuss the favourable sites for installing windmills.	04	CO3	BT2
Q.2	a	Explain the terms: camber, nacelle, cut-in speed, cut-out speed, windrose, and wind vane.	06	CO3	BT2
	b	Discuss the various types of solar radiation measurement instruments.	04	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Interpret the energy reserves of India and the potential of different energy sources.
CO 2	Measure the solar radiation parameters and performance of different solar collectors.
CO 3	Calculate different parameters of the wind turbine rotor.
CO 4	Implicit the importance and applications of geothermal and ocean energy.
CO 5	Demonstrate knowledge in the field of fuel cells and potential for power generation.



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Department of Production Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester I)

Class: TY B.Tech.

Course: Metal Cutting Theory

Course Code: PCC-PE305

Date: 09.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Questions	Marks	CO	BT
Q.1	A	5	CO1	BT2
	B	5	CO5	BT4
Q.2	A	6	CO1	BT5
	B	4	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	The students shall have the knowledge of fundamentals of metal cutting technology including the process, measurements, design and selection of various cutting tools and their industrial specifications.
CO 2	Select suitable cutting tool for performing identified machining operation.
CO 3	Design form tool for given application and conditions.
CO 4	Design single and multiple point cutting tools for given conditions and applications.
CO 5	Understand cutting forces generated during different cutting operations.



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Department of Production Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.	Course: Machine Design	Course Code: PCC-PE303
Date: 10.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Attempt all the questions.
- Attempt **any One** from **Q.1**.
- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Use of non-programmable calculator is allowed.

Q.N.		Marks	CO	BT
	Solve the following questions.			
Q.1	Briefly explain the Aesthetic Considerations and Ergonomic Considerations in design. OR	10	CO1	BT1 ,2
Q.1	Define factor of safety (<i>f_s</i>). Explain the factors affecting the magnitude of factor of safety.	10	CO1	BT1 ,3
Q.2	A feed pump of a binary vapour cycle is of centrifugal type and delivers 25 m³/min at 900 rpm against a dynamic head of 10 m . Determine the power required to drive the motor, if the pump efficiency is 85% . If the max. torque on the motor shaft is 25% more than the avg. torque, determine the diameter of the motor shaft. The permissible shear stress in the shaft material shouldn't exceed 45 MPa .	05	CO2	BT4 ,5
Q.3	A short cast iron column of hollow circular section with 300 mm outside diameter and 180 mm inside diameter carries a vertical load of 500 kN acting at a point 120 mm from the axis of the column. Find the stress induced in the column.	05	CO2	BT3 ,5

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Understand basic design procedure and the necessary design considerations.
CO2	Define and distinguish static and fluctuating loads/stresses.



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Department of Production Engg.

In-Semester Examination / Evaluation II

Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Metallurgy and Metal
Forming.

Course Code: PCC-PE301

Date: 11.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

Reg.No.....

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Solve **any FOUR questions** from questions 1 to 5.
- iii. Figures to right indicate full marks.
- iv. Paper containing 1 page.

Q.1	Explain the types of Plain Carbon Steel with their applications.	05	CO1	BT2
Q.2	Why white cast iron is hard and brittle? What are the uses of white cast iron?	05	CO1	BT1
Q.3	Explain with neat sketch Time Temperature Transformation (TTT) diagram.	05	CO1	BT2
Q.4	Explain annealing heat treatment process.	05	CO1	BT2
Q.5	Write the properties and application of copper.	05	CO1	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Classify the cast irons and their applications and heat treatment processes.
CO2	Describe powder metallurgy, its processes and applications.
CO3	Select the suitable non-destructive testing method for inspecting components
CO4	Classify different metal forming processes.
CO5	Design dies for cutting, forging, bending, drawing, piercing, and blanking.